

Ponatinib

Ponatinib is a cancer drug. It is used to treat some types of leukaemia.

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What is ponatinib?

Ponatinib is a type of targeted therapy drug called a tyrosine kinase inhibitor (TKIs).

Ponatinib is used to treat [chronic myeloid leukaemia \(CML\)](#). It is also used to treat [acute lymphoblastic leukaemia \(ALL\)](#) when the leukaemia cells have a genetic change called the Philadelphia chromosome (Philadelphia chromosome positive).

Your doctor can test your blood or bone marrow to look for the Philadelphia chromosome.

It is best to read this information with our general information about [targeted therapy drugs](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

Having ponatinib

Ponatinib comes in tablets. This means you can take it at home.

You may have it with other cancer drugs.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor (haematologist)
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment. Before starting treatment, your doctor will also check how well your heart is working.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your targeted therapy.

Your course of treatment

You usually take ponatinib for as long as it is working for you and side effects can be managed. Your cancer team will discuss your treatment plan with you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Taking ponatinib tablets

You take ponatinib once a day. Your cancer team will give you the ponatinib tablets to take home. They may give you tablets of different strengths. Always take the tablets exactly as explained. This is important to make sure they work as well as possible for you.

Make sure you:

- swallow them whole with a glass of water
- do not chew, break or crush them
- take them at the same time every day.

If you forget to take the tablets, do not take a double dose. Take the next dose at the usual time.

Other things to remember about your tablets:

- Do not eat or drink grapefruit juice while you are taking ponatinib.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused tablets. Return them to your cancer team at the hospital.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature

- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

You will have regular blood tests during treatment. If your white blood cell count is low, your doctor may stop your treatment for a short time, until your cell count increases. Sometimes they will reduce the dose of your treatment.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Indigestion or tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help.

Contact the hospital straight away on the 24-hour number if you have:

- sharp pain in your upper tummy (abdomen)
- any signs of bleeding from your back passage (rectum)
- black stools (poo)
- blood in your vomit
- vomit that looks dark and grainy like coffee grounds.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital.

Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

High blood pressure (hypertension)

This treatment can cause your blood pressure to go up. You will have regular blood pressure checks during your treatment. Tell your doctor, nurse or pharmacist if you:

- have headaches
- have nosebleeds
- feel dizzy.

If you have high blood pressure, your cancer team can give you tablets to control it. If you already take tablets for high blood pressure, tell your doctor before you start treatment.

Narrowing of blood vessels

Ponatinib can make your blood vessels narrower. This may increase your risk of having a stroke. It is important to tell your doctor if you have had any strokes or heart problems in the past. Your doctor may talk to you about the benefits and risks of ponatinib. They may also talk to you about lifestyle changes. For example, they may advise you to follow a [healthy diet](#) or to stop [smoking](#). These can lower the risk of having a stroke.

You can use the word “FAST” to remember the main symptoms of stroke.

Face:

- Your face droops on one side.
- You are unable to smile.
- Your mouth or eye may have drooped.

Arms:

- You are unable to lift both arms and keep them up because one side of your body is weak or numb.

Speech:

- You have difficulty speaking.
- Your speech is slurred or garbled.
- You are confused.

Time:

- If you notice any of these signs and symptoms, it's time to dial 999 immediately.

Effects on the liver

This treatment can affect how the liver works. You will have regular blood tests to check this. Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Raised level of fats in the blood

This treatment can raise the level of triglycerides and cholesterol. These are fatty substances in the blood. You may have a blood test to check the levels of triglycerides and cholesterol. Your doctor may refer you to a dietician for advice on diet or prescribe medicines to help.

Feeling dizzy

Ponatinib may cause dizziness. If this happens, tell your doctor or nurse. If you feel dizzy, you should not drive or operate machinery.

Fluid build-up

This treatment can cause a build-up of fluid in the body. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called peripheral neuropathy. You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the dose of the drug.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Hair loss

You might notice that your hair slowly gets thinner. But you are very unlikely to lose all the hair from your head. Your hair can also become more brittle. Your nurse can give you information about [coping with hair loss](#). Your hair will usually go back to normal after treatment ends.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Effects on the eyes

This treatment may affect your eyes or eyesight. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- your experiencing dry eyes
- you have pain in your eyes
- you notice any change in your vision
- you notice any unusual eye movements.

Effects on the lungs

This treatment can cause a build-up of fluid in the lining of the lungs. This is called a [pleural effusion](#). Contact the 24-hour number if you have any of these symptoms:

- breathlessness, even when you are resting
- a cough
- chest pain.

Effects on the heart

This treatment can affect how the heart works. You will have [tests](#) to check how well your heart is working. These will be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing

Erectile dysfunction

You may have difficulty getting or maintaining an erection while taking ponatinib. Talk with your cancer team if you are worried about this.

Anxiety

You may feel anxious during this treatment. Talking to family and friends about how you feel might help. If anxiety lasts for more than a few weeks, tell your doctor, nurse or pharmacist. They can talk to you about different ways to manage anxiety.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Thyroid changes

Ponatinib can affect your thyroid gland. You will have regular blood tests to check this.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)

- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

- a headache that does not get better
- drowsiness or confusion
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Hot flushes and sweats

Hot flushes are a common side effect of this treatment. During a flush, your neck and face may feel warm. If you have white skin, your skin may look red. If you have black or brown skin, your skin colour may deepen. Flushes can last up to 5 minutes. You might also have sweats or feel anxious or irritable during a flush.

Here are some things you can do to reduce the effects of hot flushes:

- Wear natural fabrics, such as cotton.
- Wear layers so you can remove clothes as needed.
- Use cotton sheets and layers of bedding you can remove.

- Try cooling pads or pillows to keep you cool.
- Keep rooms cool or use a fan.
- Have cold drinks and avoid caffeine and alcohol.

You might have fewer hot flushes as your body adjusts to this treatment. You could try therapies to help you [cope with hot flushes](#), such as:

- [talking therapies](#) like cognitive behaviour therapy (CBT)
- controlled breathing exercises
- yoga
- acupuncture.

If they do not improve, talk to your doctor. Certain drugs can help to improve hot flushes.

Flushes usually stop a few months after treatment ends. But some people continue to have them.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other important information

Hepatitis B reactivation

If you have had a liver infection called hepatitis B, this treatment can make it active again. Your doctor, nurse or pharmacist will talk to you about this. You may have tests for hepatitis B before, during, and after treatment.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Lactose

This treatment contains lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help